



# Detailed Backend Documentation

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# Detailed Backend Documentation

[Build Documentation](#) and [Design Documentation](#) give a snapshot of the backend. Separately, the `Documentation/` folder in the main repository holds a deeper, actively-maintained documentation pass over the backend ( `Backend/api` , `Backend/workers` , `Backend/shared` ) — how it's designed, what each file does, and how to run/use it. This page explains what's in that folder and links straight to each file.

**Start here:** [Documentation/README.md](#) is the master index — it links out to everything below plus the team's Git workflow guide. Everything on this wiki page is a shorter version of that same index, adapted for the wiki.

Four parts:

| Part                                  | Answers                                   |
|---------------------------------------|-------------------------------------------|
| <a href="#">Architecture</a>          | How is it designed? (diagrams)            |
| <a href="#">Code Components</a>       | What does this specific file/function do? |
| <a href="#">Running the Project</a>   | How do I try it or run my own copy?       |
| <a href="#">API Integration Guide</a> | What's the exact shape of this endpoint?  |

## Architecture

`Documentation/architecture/` — the "why" and "how it flows" layer, with Mermaid diagrams. Covers both AI pipelines, the API service, the literature-ingestion job, and the GCP deployment topology.

| File                                                           | Covers                                                                                                   |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <a href="#">README</a>                                         | System context diagram, tech stack, deployment topology at a glance                                      |
| <a href="#">01 — Extraction &amp; Knowledge Graph Pipeline</a> | Upload → Gemini multimodal document analysis → Graphiti/Neo4j knowledge graph → Cypher export            |
| <a href="#">02 — Question Answering Pipeline</a>               | Chat/voice RAG: contextualize → retrieve (knowledge graph + research papers) → answer                    |
| <a href="#">03 — API Service Architecture</a>                  | FastAPI app structure, auth, the document upload lifecycle, middleware, error model, security boundaries |
| <a href="#">04 — Literature Ingestion Pipeline</a>             | The standalone scraper job that feeds the research-paper corpus chat reads from                          |
| <a href="#">05 — Infrastructure &amp; Deployment</a>           | Cloud Run topology, IAM, data stores, Pub/Sub, Terraform layout                                          |

This is where the pipeline actually described in this wiki's [[Design Documentation]] is current — e.g. the extraction pipeline moved from Google Document AI OCR to Gemini multimodal analysis + a fixed medical entity/edge schema since that page was last updated.

## Code Components

Documentation/code-components/ — a per-file reference catalog of Backend/api , Backend/workers , and Backend/shared : purpose, key exports, and non-obvious notes for every module. Use this once you already know which pipeline you're in (from Architecture above) and need the exact function.

| File                                   | Covers                                                                        |
|----------------------------------------|-------------------------------------------------------------------------------|
| <a href="#">README</a>                 | Index + directory map                                                         |
| <a href="#">01 — Shared</a>            | shared/ — Firestore models, repositories, GCS/Pub-Sub/Firestore infra clients |
| <a href="#">02 — API Core</a>          | api/main.py , auth.py , errors.py , middleware.py , chat.py , voice.py        |
| <a href="#">03 — API Chat Pipeline</a> | api/chat_pipeline/ — all 9 modules behind contextualize → retrieve → answer   |
| <a href="#">04 — Workers Core</a>      | workers/main.py + workers/connections/ (Gemini, Neo4j, GCS clients)           |

| File                                  | Covers                                                                                               |
|---------------------------------------|------------------------------------------------------------------------------------------------------|
| <a href="#">05 — Workers Pipeline</a> | workers/pipeline/ — orchestrator, LLM extraction, graph building, Cypher export, deprecated OCR path |

## Running the Project

Documentation/running-the-project/ — task-oriented, not reference. Splits into the two audiences an open-source project actually has: people who just want to try the already-deployed system, and people who want to run their own copy.

| File                                           | Covers                                                                                                                                                                        |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">README</a>                         | Which of the two paths below you want, and what the project's compute topology actually is                                                                                    |
| <a href="#">01 — Using the Deployed System</a> | Zero-setup path: curl walkthrough against the already-deployed backend — sign up, upload+extract a document, chat against it                                                  |
| <a href="#">02 — Running Locally</a>           | Full local dev setup: prerequisites, credentials, environment variables per service, running <code>api + workers</code> with <code>uvicorn</code> , Docker, local smoke tests |

This supersedes the "Backend Build Documentation" section of [[Build Documentation]] for anything beyond a one-line reminder — in particular the environment-variable tables here are sourced directly from the current code rather than `Backend/.env.example`, which is itself out of date.

## API Integration Guide

Documentation/api-integration-guides/ — the full field-by-field endpoint reference: exact request/response shapes, every error code, the document status state machine, and the polling pattern for extraction progress.

| File                                              | Covers                                                       |
|---------------------------------------------------|--------------------------------------------------------------|
| <a href="#">Document API FE Integration Guide</a> | Every endpoint — auth, documents, <b>chat</b> , <b>voice</b> |

This is also linked from [Accessing & Visualising the Knowledge Graph](#) (the `graph_query` / `entities_query` fields it points at are documented here).

# How this relates to the rest of the wiki

- [User Documentation](#) — what the *app* does, for someone who isn't reading code at all.
- [Build Documentation](#) / [Design Documentation](#) — this wiki's own architecture/setup write-up, useful as a faster-to-skim summary; treat the `Documentation/` folder above as the source of truth when the two disagree, since it's checked against the current code more recently.
- [Accessing & Visualising the Knowledge Graph](#) — a focused how-to that the Architecture and API Integration Guide docs above both link into.

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